



JÖNKÖPING UNIVERSITY

*School of Engineering*

# Designing Loading Screens to Improve UX

Comparing Static, Animated and Interactive Interfaces

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## Abstract

Reaching the best user experience (UX) using a wide range of visual, psychological, and technical tricks and skills is the main aim of designers. However, despite a great design, situations can occur when the overall experience on the website can still be ruined. A frequently occurring example is the unexpected loss of internet connection, which forces the users to wait for an uncertain amount of time. It is a critical design challenge, since delays can lead to user frustration and even to task abandonment. This thesis investigates how different types of loading screen designs—static, animated, and interactive—impact user frustration, engagement, and perceived wait time. Drawing from theories in emotional design, colour psychology, and UX, a within-subjects experiment involving 30 participants was conducted using custom-designed e-commerce website prototypes. Each participant experienced all three loading screen types (appearing for 10 seconds), and their responses were measured through both quantitative (Likert scale) and qualitative (open-ended interviews) methods. The results showed that interactive loading screens significantly reduced frustration and increased engagement. The perceived waiting time was not significantly different, but the interactive screen was perceived to be slightly shorter compared to static and animated counterparts. Animated screens moderately improved the experience, while static screens were least effective. These findings highlight the value of interactivity and emotional engagement in UX design, offering practical implications for developers seeking to improve UX and satisfaction during inevitable wait periods. The study highlights the importance of designing loading experiences not just as technical necessities but as opportunities to enhance user perception and trust.

*Keywords: User Experience (UX), Emotional Design, Loading Screens, Interactive Design, User Frustration, User Engagement, Perceived Waiting Time*

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# 1 Introduction

Nowadays, the role of user experience (UX) design, particularly on e-commerce websites, is to create a smooth shopping experience for users. An experience that can elevate customer satisfaction, making them revisit a well-designed website. A website that can create a more enjoyable, trustworthy environment (Faisal et al., 2017). When it comes to online shopping, users crave a fast-paced experience and have no patience when it comes to waiting (Kirilka et al., 2012). Multiple studies (Cheng et al., 2024; Wang et al., 2021) have shown when users experience long waiting times, they become frustrated, tend to abandon the website, and lose trust in the platform. Online shopping has become part of people's daily routine, where long waiting times can have a negative impact on the customers and their approach towards the website's image (van Riel et al., 2012). These are critical effects that can lead to a bad reputation for the brand's image and even potential drops in sales for the company.

E-commerce websites have advanced in higher quality resolutions and new features, yet little research has been done on a good solution to waiting time (Nah, 2004). There are already a few loading screens created, such as the progress bar or the infamous offline "Dinosaur Game" by Google. Nevertheless, there is a lack of research exploring the loading screen best suited for websites during waiting time. Even though loading screens are often designed to lessen the level of frustration, their effectiveness has yet to be fully proven. Recent studies have mainly focused on simple loading screen types, yet little research has been done comparing these types to determine which is best suited for UX. This thesis emphasises the type of loading screen that can solve the issue of long waiting times and the importance of user satisfaction. More specifically, it focuses on the three screen types, namely static, animation, and interactive. These prototypes are tested to understand which of them performed the best as a result of the users' feedback (Söderström et al. 2018). For example, in a study by Persson (2019), the impact of five different types of loading animations was examined. The results showed that loading screens with animations were preferred over no animation. This highlights the impact animation can have on perceived performance. However, to our knowledge, the UX of a mini-game as a loading screen remains unexplored. Therefore, this research aims to compare the traditional screen approaches (static and animated) with an interactive game in an e-commerce setting. This approach allows prototypes to be tested and compared based on user preference for e-commerce loading screens.

This thesis begins with the background explaining the key concepts that are relevant to the study. Furthermore, related works are introduced to establish the context of the thesis. The motivation is later addressed with the problem statement, followed by the research questions, the scope, and the limitations. In addition, the method and implementation discuss the research and product design, the methods used to test them, and the sampling procedure. To conclude the thesis, the expected results are addressed to further understand the reasoning behind this study and the potential impact the different loading screens can have on UX.



## **2 Background**

In this section, every key concept is defined and analysed for a better understanding of this thesis. It begins by exploring UX and user frustrations, including concepts such as UX design and enhancing user interactions. It continues with an overview of emotional design and colour theory, and concludes with an in-depth explanation of user frustration, user engagement, the psychological effects of waiting time, and how design can help address these challenges.

### **2.1 User Experience Design**

Creating a good UX involves deliberately thinking about design choices (Garrett, 2006). Usability testing can be a way to enhance a specific task, with the need to provide an outcome of pleasure to the user when it comes to this study. An experience can be perceived in different ways, such as subconscious, fluent, and automatic, or one that requires focus and concentration (Battarbee et al., 2005).

According to Wang (2024), excellent UX design relies on the user's perception, their needs and emotions. This is achieved through a pleasant and user-friendly experience. The main objective of UX design is to be able to design an interface that not only engages with the user but can also evoke emotional feedback.

#### **2.1.1 Emotional Design**

Emotional design plays a vital role in the context of UX as it can affect the way a person feels. It has been mentioned that visual aesthetics influence the way a website is perceived as well as the users' feedback, like satisfaction and enjoyment (Lavie & Tractinsky, 2004; Moshagen & Thielsch, 2010; Schenkman & Jönsson, 2000; Tuch et al., 2010; van der Heijden, 2003, cited by Heidig et al., 2015). Moreover, researchers have found that positive emotions can improve motivation, problem-solving skills and creativity (Chen & Wang, 2011; Um et al., 2012, cited by Heidig et al. 2015). With a design that can capture the feeling of satisfaction and joy, the right colours are just as important to complete the prototype.

#### **2.1.2 Colour Theory**

According to Jonauskaite et al. (2020), colours have a huge impact on the users' feelings and emotions. Heidig et al. (2015) state that to have positive emotional responses, colour combinations, along with pleasing colours, influence the responses. Colours that generate the feeling of quiet and calmness are cool colours such as blue, green, and purple, while colours that are perceived as appealing and bold are warm colours like red, yellow, and orange (Ballast, 2002, cited by Kaya, 2004). It is proven that the colour blue reveals a great number of emotional and positive feedback with feelings of tranquillity, relaxation, peace, joy, and comfort while having fewer negative emotions, including sadness and depression (Kaya, 2004). More about the



reasoning behind choosing these specific colours will be mentioned in the Methods and Implementation section later in the research.

## **2.4 User Frustration and Time Perception**

### *User Frustration*

User frustration can be triggered by loading breaks and long waits (Heidrich et al., 2020). Lazar et al. (2006) explain frustration as causing personal annoyance and boredom, leading to a decrease in participation and self-confidence as well as work efficiency. Long waits can create a negative experience since users subconsciously translate the actual waiting time as longer than it actually is (Heidrich et al., 2020). Meaning, users can perceive the waiting time to be longer than in reality. Thus, by adding design features, it can reduce negative emotions, making the perception of waiting time feel shorter.

### *Time Perception*

Hertzum et al. (2023) conducted a study resulting in participants expressing their frustrations during the waiting time, which negatively influenced their capability to complete an online task. Thus, showing that during long waiting times, frustration is increased, resulting in slow task achievement. It has been studied that directing users' attention to different situations without the visual content of time makes the long waiting time feel shorter and more positive when additional content is present, as opposed to focusing on the time (Hong et al., 2013). Additionally, the most commonly acknowledged aspect of the user's perceptions of computer frustration was the delayed and unresponsive systems (Hertzum et al., 2023). In relation to this thesis, user frustration was measured with users reacting to the prototypes and identifying whether it influenced a positive or negative experience. Taylor (1994) proposes that filling time can help lower the users' boredom, anxiety, and distrust. Therefore, it makes the waiting time more tolerable and can ease the user's frustration. Long waiting times with additional content keep the users distracted from the perception of time, in contrast to information that signals the passage of time to the user (Lee et al., 2017). Theories behind the perception of time can be a tool for handling online waits since it informs the way people determine the passing of time and ways of improving the waiting time.

### 3 Related Work

No studies were found comparing static, animated, and interactive loading screens. However, researchers have compared other kinds of loading screens (e.g., Wang et al., 2021, Cheng et al., 2024).

Wang et.al. (2021) examined how different types of initial loading feedback (ILF) in mobile applications influence users' mental states and their tolerable waiting time (TWT). The authors tested four ILF conditions: a blank screen (Null), a spinner (Circle), a static logo, and a combined logo with a slogan. This serves as a good comparison to the static and animated conditions of this thesis. Their experimental setup involved 90 participants and used a within-subjects design.

Participants were asked to rate their mental state using emotional (pleasure, arousal, dominance) and cognitive (attention, perceived speed) dimensions, as well as to report how long they would be willing to wait in each scenario. The key finding was that loading screens with meaningful content, such as logos and slogans, improved UX, increasing perceived credibility, attention, and tolerance to waiting. Conversely, the Null (blank) and Circle (spinner-only) designs were rated lowest across most dimensions, especially in terms of trust and perceived wait duration. Interestingly, although the spinner was a commonly used loading indicator, it resulted in shorter TWT and less user satisfaction than logo-based designs.

The central research question in Wang et al.—whether certain designs reduce user frustration and extend tolerable wait times—mirrors the aim of this thesis, which seeks to evaluate whether static, animated, or interactive loading screens can effectively manage user expectations and reduce frustration in a web-based environment. However, while Wang et al. (2021) focus on static and passive elements, the present thesis builds on this by adding interactivity as a key variable. Their findings lay the groundwork for understanding the psychological impact of loading design and support the idea that visual richness and meaning can reduce the negative effects of delay. This provides a strong conceptual foundation for this thesis.

Another relevant paper by Cheng et al.(2024) similarly studies how interactivity and visual design in loading screens affect users' emotional experience, time perception, and preference. Cheng et al. (2024) also compared three versions, but they used a 3×3 factorial design to explore user responses to three loading screen types (a passive animation and two interactive variations) across three different waiting times (2s, 5s, and 10s). During the tests, Cheng et al.(2024) also use a within-subject design for comparing the screen types, but a between-subjects design for waiting durations. Likert scales were used to measure emotions and preferences. They found that adding interactive or dynamic elements to loading screens can reduce user frustration and enhance engagement for longer durations. Cheng et al.(2024) also discusses colour as an emotional and cognitive factor, which encouraged the researchers of this paper to choose the

colours of the loading screen, taking into consideration colour theory principles described in the background section.

## **4 Motivation and Aim**

### **4.1. Problem Statement**

The field of UX aims to maximise the experience of the users. Designers use a wide range of visual, psychological, and technical tricks and skills to reach it. However, situations can occur when, despite a great design, the overall experience on the website can still be ruined. A frequent example is the sudden loss of internet connection, which forces the users to wait for an uncertain amount of time. The motivation behind this study is to test out which of the three prototypes is most effective in reducing frustration and can have an impact on users' satisfaction during waiting time. The study focused on the users' experience and emotions and determined whether loading screens are necessary in situations with poor internet connection.

Research shows how users tolerate waiting time (TWT) differently based on feedback. Without any loading feedback, TWT is as low as 5–8 seconds, whereas interactive feedback can extend TWT up to 38 seconds (Nah, 2004 as cited in Website Optimisation, 2008). When the response of the website is delayed without meaningful engagement, users are more likely to abandon tasks or develop negative perceptions of the system (Krishnan & Sitaraman, 2012, as cited in Cheng et al., 2023).

Studies also emphasise the importance of maintaining user engagement during wait times (Cheng et al., 2024; Martín-Peña et al., 2023; Website Optimisation, 2008). Research suggests that interactive elements, such as small loading screen games, can increase user engagement and improve perceived wait times (Cheng et al., 2024). According to participants' test results, interactive design is more successful since it is viewed as faster during the waiting time and is enjoyed by more (Hohenstein et al., 2016).

Despite these insights stated in this section and in the related works section, the researchers of this paper have not found studies that have directly compared static, animated, and interactive loading screens in a web context to determine which is most effective at minimising frustration and enhancing UX. Thus, the current research addresses this gap by evaluating user responses to three loading screen types under controlled, realistic, web-based conditions.

### **4.2. Purpose and Research Question(s)**

Following the gap stated above, this study seeks to explore how different loading screen interfaces—specifically static, animated, and interactive formats—affect UX during long waiting times (10 seconds) in an e-commerce environment. By analysing emotional responses and time perception, this research aims to provide evidence-based recommendations for improving UX during loading delays.

The research questions guiding this study were:

***RQ1: How does the user interface during long loading times influence the (a) frustration level, (b) engagement level, and (c) perceived waiting time?***

***RQ2: Is perceived lower time something that predicts frustration?***

***RQ3: Which type of interface (static, animated, interactive, etc.) has the most positive effect on users' experience during long loading times?***

By answering these questions, the study aims to provide valuable insights for designers seeking to build more resilient and user-friendly digital environments. Specifically, it focuses on improving loading screen design to enhance user experience, boost engagement, and reduce frustration during waiting times.

### **4.3. Scope and Limitations**

The scope of the user tests is an environment where the user is looking at an e-commerce website. All the loading screens appeared for ten seconds because the research did not have the capacity to test more variables. There was also no time to test more colour combinations, characters, or any other design variation for the loading screens. Thus, the design was consciously limited to versions that look as similar to each other as possible to reduce the number of small details, which could influence users' impressions.

The subjects of the research were the students enrolled at Jönköping University, specifically the age group of youth; thus, claims cannot be made about other populations. The reason behind the chosen age group was about the extensive experience in digital technology; the younger generation, as Prensky (2009) referred to as 'digital natives,' has been compared to the older generation. Even though the subjects come from several different countries, there are not enough participants for each country to make a relevant representation for each country or region.

Perhaps the primary limitation of this paper is that the researchers needed to stick to one geographical place and a relatively short time frame. This has led to the limit of not having the capacity to collect a large group of participants. It allowed only the use of a within-subjects method, which has the typical disadvantage of the carry-over effect (Greenwald, 1976). The within-subjects design also causes another weakness, the potential serial-position effect explained in the background section. Furthermore, unlike Cheng et.al (2024), who tested three different time variables, this paper only had the capacity for testing one time length (10 seconds).

Furthermore, it is important to note that the information users give during the post-test interviews is only self-reported. For example, when the users are asked for their level of frustration, their emotions are not measured objectively in any way, so researchers can only rely on the data that the users claimed to have.

## **5 Method and Implementation**

### **5.1. Research Design**

A controlled study with repeated measures was conducted. Participants interacted with three prototypes: static, animated, and interactive. They were presented in a randomised order. According to Aleferes (2012), randomisation is essential for collecting unbiased estimates. Thus, this approach ensures that the order in which the prototype was tested didn't affect the user's response, thereby enhancing the reliability of the results.

The within-subjects design was chosen because it allows direct comparison of user reactions to different loading screen designs, controlling for individual variability among participants. This concept was also used by Wang et al. (2021), who successfully employed within-subject methods to effectively evaluate user emotional responses, perceived waiting times, and frustration levels when exposed to different loading screens.

The main idea for the loading screen design was to showcase a calm yet playful environment. It is known by researchers that design can have an impact on users' satisfaction (Söderström et al. 2018). All three screens were designed to have an underwater theme with a small fish being the centre of attention. As mentioned in the background, both warm and cool tones were chosen to portray a sense of calmness (blue) for the screen's background and a fun and joyful essence (yellow and orange) from the fish to grab the users' attention. Moreover, features such as the text "No Internet Connection" were added at the bottom to indicate the message that there was an issue involving the internet. Both the animated and interactive screens had additional bubbles and rocks added as an extra detail to keep the customer entertained and maintain engagement during a long waiting time.

Nowadays, users often shop online for specific items yet may encounter internet issues that can impact their mood in a negative way, leading to an unpleasant experience and potentially exiting the website altogether. For that reason, an e-commerce website was chosen as the platform to test the prototypes in a randomised order, as it reflects the effect of different loading screens in realistic, everyday online shopping. After each prototype testing, all participants were interviewed using Likert Scales (see Figure 5.1.2.2) and in-between questions (see Figure 5.1.2.3) as a way to acquire immediate impressions. Conducting the interviews after each

prototype, as opposed to the end, ensures the users' thoughts are still fresh, reducing the risk of any inaccuracy. A post-test interview was also conducted with open-ended questions (see Figure 5.1.2.4) to finalise the data and get an overall clear result from all prototypes. This gives the users a chance to reflect on their experiences. To test out the prototypes, it was decided to place them within the context of an e-commerce website, as it best showcases the designs. Users were asked to search for a product on the website. They later encountered an issue with their internet connection; therefore, one of the loading screens appeared. This determines if adding a design to the loading screen impacts their experience positively or simply has no effect during the waiting time. By comparing the reactions of the users, it can be identified which of the three loading screens is most successful in this situation.

### **5.1.1. Prototype Designs: (a) Loading Screens & (b) E-Commerce Website**

Studies have shown that the design of a loading screen can affect the users' experience and emotions. It was shown to decrease the perceived wait time (Hohenstein et al., 2016). It was tested that the type of animation used for loading screens can influence the perceived performance (Persson, 2019). The loading screens were intentionally designed to be as similar as possible between conditions to avoid other design factors than the ones investigated from having an influence on the factors investigated.

#### *Loading Screens*

##### **Static Prototype:**

For prototype 1, a static fish was designed in Adobe Illustrator (see Figure 5.1.1.1) using colours that have a positive effect on people since it has bold colours of yellow and touches of orange (Ballast, 2002, cited by Kaya, 2004). The fish was chosen as the main illustration as it portrays a friendly and cheerful appearance. Its happy expressions were designed to create cuteness and a positive feeling, making the waiting time and experience more engaging and fun for the users.

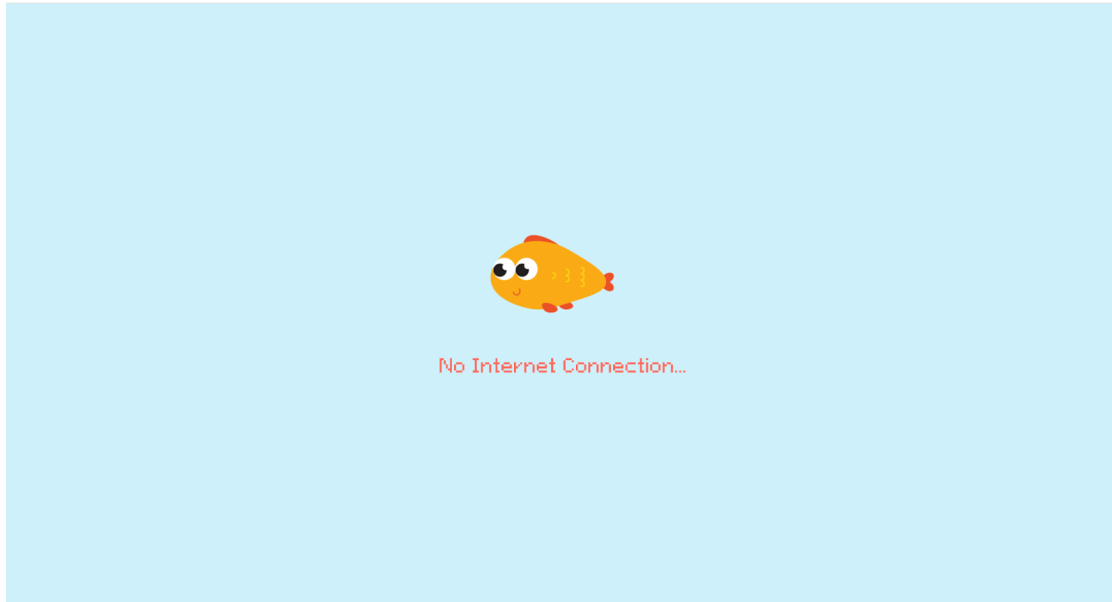


Figure 5.1.1.1: *Static Loading Screen*

**Animated Prototype:**

Using the same design, however, for prototype 2, the fish is animated using the tools in Adobe After Effects, where the fish moves upwards and downwards, blowing bubbles to keep the user even more engaged.

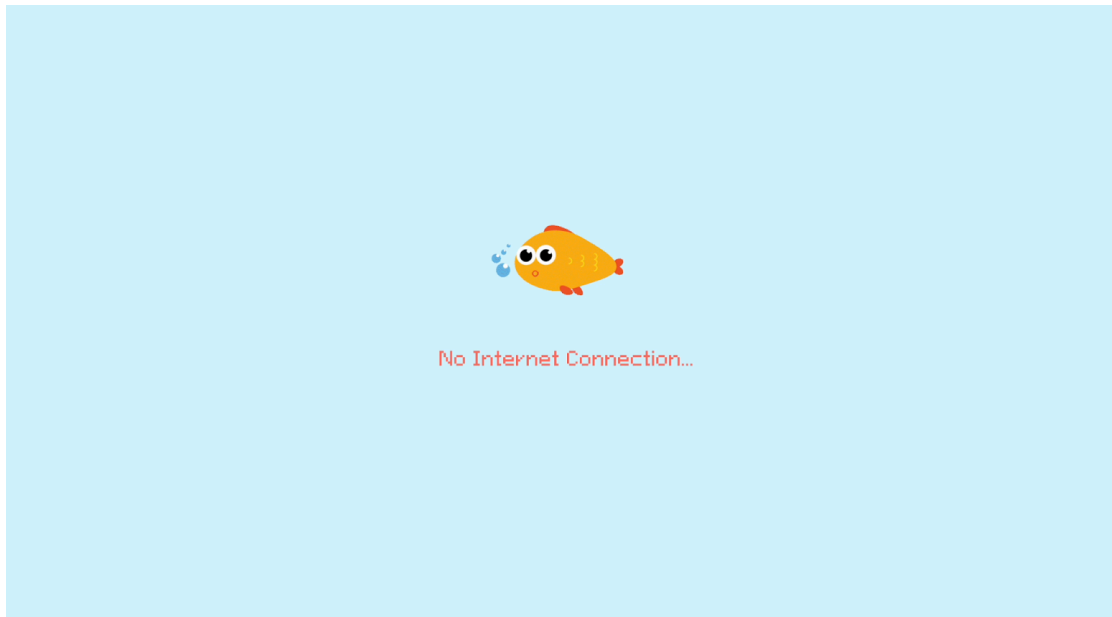


Figure 5.1.1.2: *Animated Loading Screen*

**Interactive Prototype:**

Prototype 3 is the interactive design that was built on Visual Studio Code, a cross-platform editor (Johnson, 2019), using JavaScript as the main programming language. The game was coded to



create falling bubbles and rocks, where the goal was for the users to collect as many bubbles as they could while avoiding any collisions with the rocks. To navigate the fish, users had to move their mouse to avoid the collisions and collect as many bubbles as they could. It was discussed that developing a minigame would help the user to stay engaged and entertained during the loading screen, reducing potential frustration and enhancing the UX on the website. According to participants' test results, interactive design is more successful since it is viewed as faster during the waiting time and is enjoyed by more people. (Hohenstein et al., 2016)

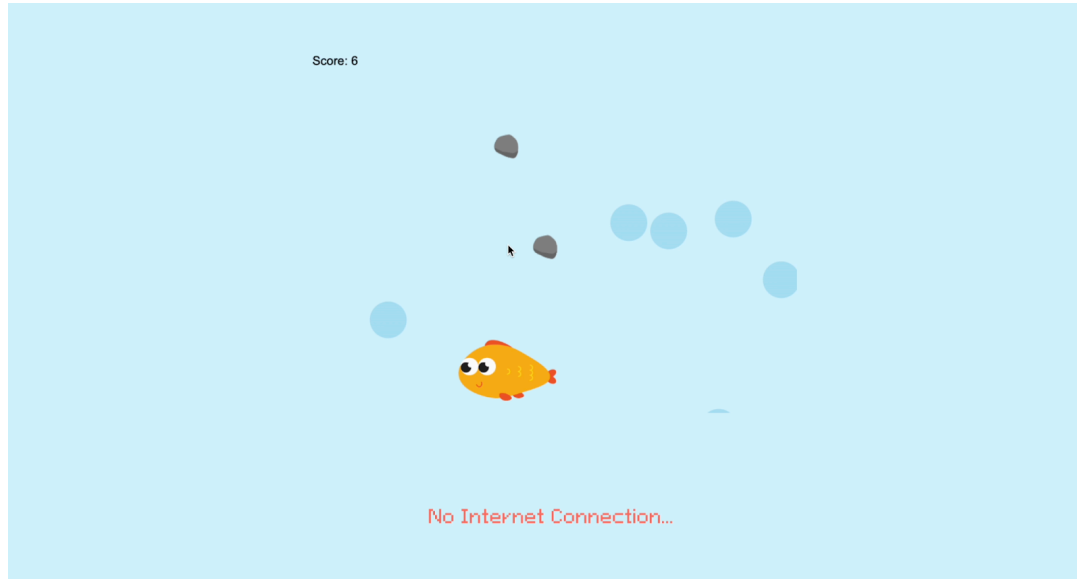


Figure 5.1.1.3: *Interactive Loading Screen*

### **The E-Commerce Website**

As previously mentioned, for the three prototypes, a situation had to be created where the users are using a website, and the internet connection just suddenly breaks. JavaScript was used to create a custom-coded website that needed to include the loading screens uniquely.

A website was needed where users could perform tasks and go through the site, and click on several buttons. It was assumed that concentrating on achieving these tasks would make them more surprised when the loading screen suddenly appeared. An e-commerce website was chosen for this goal.

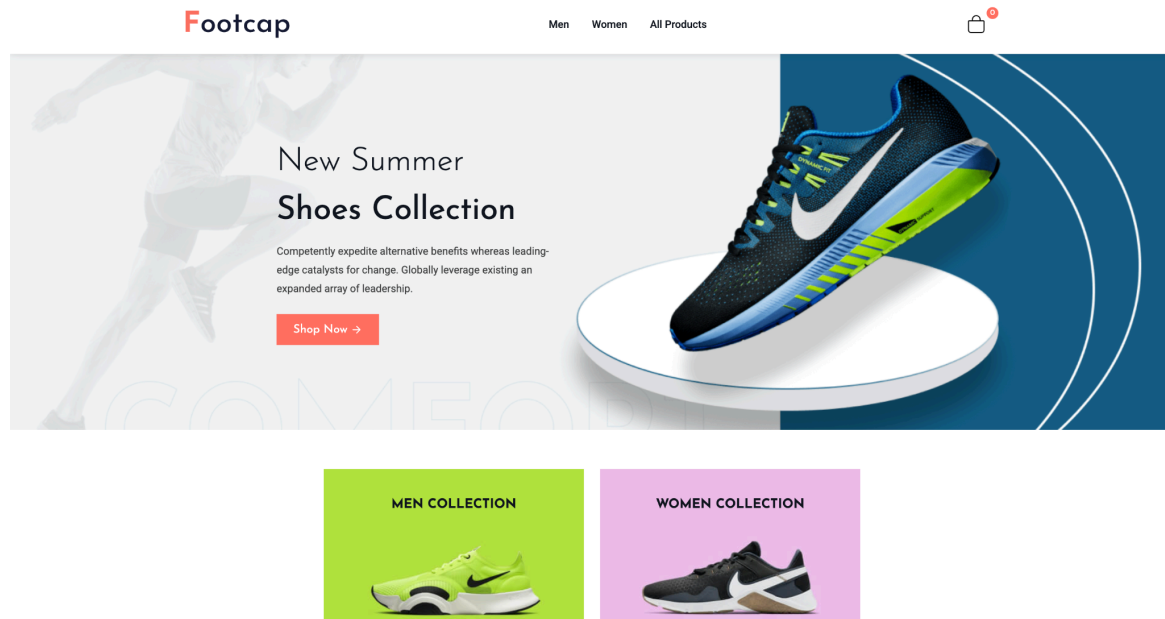


Figure 5.1.1.4: *Landing Page of the E-Commerce Website*

Since the user-friendliness and the front-end design of the website were not in focus, the website is quite simple in terms of structure and visuals. It has a landing page from which users can jump to the men's and women's collections. Through the navigation bar, the landing page, men, women, all collections, and the cart can be reached. The number above the cart icon is updated whenever a product is added to the cart. From the cart page, you can continue with creating an account by filling out a form and then buying the product at the end.

Simulating a realistic internet connection loss was a significant design challenge. The system was designed so that users would trigger the appearance of the loading screen through their interactions. The participants cannot see it on the client side, but some buttons or links were coded to redirect to the loading screen site and, after ten seconds, automatically back to the site where the interaction was made. Since the goal was for the users to have the least expectation when the loading appears, it needed to be placed on different parts of the website. As a base, the same website code was used, but modifications were needed for each prototype by putting the code of the loading screen in different places. The tasks and codes were also created in a way that the user „calls” the loading screen twice per prototype. Each prototype is connected to a different task, so a clear message saying “Thank you for shopping” also appears at different stages on each version. Hence, achieving a realistic internet-connection-loss was accomplished by coding the prototype in a way that the “internet-loss” screen appears in different places on the website, so it appears unexpectedly.

## Prototype 1:

The loading screen is *static*, using a simple .png as background. It appears when the user clicks on the button „Add to Cart”. The user will call it twice since it is asked to put two products in the basket.

### Prototype 2:

The loading screen is *animated*, using a GIF file. It appears when the user clicks on the product and the „Buy” button.

### Prototype 3:

The loading screen is a simple JavaScript-coded *interactive* mini-game described earlier in detail. It appears when the user presses the buttons „Proceed to Checkout” and „Create an Account”.

## 5.1.2. Between/Post-test Interview Design

Before starting the tests, participants were informed about the instructions of this experiment, making sure they were comfortable and understood what they needed to do. The interviewers had a list of questions (see the questions with all the answers in Appendix A) to gather the participants’ feedback about their experience. It began by collecting user information, such as age, gender, and country of origin (see Figure 5.1.2.1).

|                          |
|--------------------------|
| <b>USER INFOS</b>        |
| Age: _____               |
| Gender: _____            |
| Country of Origin: _____ |

Figure 5.1.2.1: *User Info Questions*

Right after each prototype, a semi-structured interview with in-between questions was conducted with the goal of measuring how frustrated and engaged participants felt when each loading screen appeared. However, since frustration and engagement itself could not be assumed, participants were presented with a neutral statement and asked to rate their level of agreement on a Likert scale from 1 (Strongly Disagree) to 10 (Strongly Agree) to indicate whether they felt frustrated.

Please rate your agreement with the following statements using a Likert scale ranging from 1 (Completely Disagree) to 10 (Completely Agree).

1. I felt **frustrated** when the loading screen appeared.  
1 (Completely Disagree), 2, 3, 4, 5, 6, 7, 8, 9, 10 (Completely Agree).
2. The loading screen made me **engaged**.  
1 (Completely Disagree), 2, 3, 4, 5, 6, 7, 8, 9, 10 (Completely Agree).

Figure 5.1.2.2: *Likert-Scale Questions*

The last two in-between questions focus on asking users how long the time it took during the process and their general thoughts about the prototypes (see Figure 5.1.2.3). This was to gather data regarding the perception of time. Additionally, after all the prototypes had been tested, a general post-test interview was conducted, which consisted of four open-ended questions (see Figure 5.1.2.4). This gave the chance to get a deeper understanding of how the users felt during the process, as well as general feedback that could be used for future studies and improvements.

3. How **long** do you think the loading screen lasted? (in seconds) \_\_\_\_\_
4. What did you **like and dislike** about this loading screen?

Figure 5.1.2.3: *In-between Questions*

**GENERAL QUESTIONS**

13. Which loading screen prototype gave you the **best experience** and why?
14. In your opinion, which loading screen made the wait feel the **shortest**? Why?
15. Did the loading screen's design enhance your **overall experience**? Please share your thoughts.
16. In a **real-life** situation where your internet connection is lost, which of the three loading screens would you prefer to see, and why?

Further Notes or thoughts they shared: \_\_\_\_\_  
\_\_\_\_\_

Figure 5.1.2.4: *Open-Ended Questions*

## **5.2. Sampling Procedure**

30 participants aged between 19 and 32 were chosen for this study. This sample size is particularly suitable for interviews, as recommended by Marshall et al. (2013). The people were students at Jönköping University who were asked to participate in this experiment with their consent. More specifically, these students were enrolled in international courses at the university, providing a diverse range of perspectives and feedback coming from different cultural backgrounds.

### **5.2.1 Procedure**

This part of the thesis describes the different methods that were used to collect the data, and whether it was able to answer the research questions. The methods that were chosen for this study were both qualitative and quantitative, including conducting a controlled experiment, user-testing, and interviews.

## **5.3. Data collection**

The study was set in a booked room at Jönköping University, where users were given a laptop with an e-commerce website selling shoes. Participants were asked to browse the site and buy a pair of shoes, reaching the checkout page. During their task, a loading interruption occurred with the intent to present one of the three prototypes (static, animated, or interactive). After each prototype, the users were asked follow-up questions: two using a Likert scale to measure the perceived frustration and engagement level, and two open-ended questions that allowed them to share their thoughts and feelings about the experience.

Once all three tasks had been completed, users were asked four additional questions about their overall experience, preference, and any suggestions they may have for improving the prototypes. Their responses were documented both on paper and typed into a laptop. Participants were encouraged to answer as detailed as possible, enhancing the qualitative data.

## **5.4. Data Analysis**

The data was analysed using both statistical and thematic analysis to gain a better understanding of the users' experience between the three prototypes.

The statistical analysis focused on the quantitative data through the Likert scale, ranking the frustration and engagement levels. Each of the participants' responses was collected, and mean values and standard deviations were calculated for the static, animated, and interactive prototypes. To compare the relationship between the frustration level and perceived time, a Pearson correlation analysis was conducted. This approach made it easy to understand whether the high frustration level had any relation to the long perceived waiting time. To visually

compare the three prototypes, a table was made by adding the participants' testing order, their rating of the frustration and engagement levels, along with the perceived time. This was to get a clear analogy between the three prototypes. The purpose of this analysis was to identify any similarities and differences between the prototypes, which prototype caused the least frustration, felt the most engaging, and perceived the shortest waiting time. By using this method, the response pattern that differs between prototypes was identified, showcasing the most effective design approach.

The thematic analysis was conducted on the qualitative data from open-ended questions. These questions were evaluated and used as references. They were carefully viewed, and the most commonly used phrases and expressions related to the prototypes with UX were identified. The users' responses are presented as either quotes or paraphrases throughout the report, supporting specific findings. This gives a clearer understanding of *why* users preferred one of the prototypes over the other two. It demonstrates a deeper understanding of their choice of favoured loading screen through detailed information that numerical data cannot fully capture.

## **5.5. Validity and Reliability**

To strengthen **validity**, the study used a within-subjects design where each participant interacted with all three prototypes (static, animated, and interactive) in a randomised order. This helped control for individual differences that could otherwise bias results. Immediate post-test interviews and Likert-scale questions ensured that participants' feedback was captured while their impressions were still fresh, reducing the risk of recall accuracy. Furthermore, the experimental setup, based on realistic e-commerce tasks, was carefully designed to simulate real-world scenarios and user behaviour, enhancing ecological validity.

To create **reliability**, every effort was made to ensure that the measurement instruments—the Likert-scale questions and interview questions—were applied equally across all participants. Keeping the task and waiting time (10 seconds) the same across participants for each prototype further enhanced reliability.

In conclusion, the study achieved a high degree of validity and reliability through careful method selection, real-world task simulation, and randomised testing procedures.

## **5.6. Considerations**

Several considerations were acknowledged during the process of the study. This included both scientific and ethical considerations that had relevance in the means of our research.

A scientific consideration for this study was the use of a within-subject experimental design. This allowed the user to compare the three prototypes and give distinct feedback. While this

structure ensured diverse opinions and observations, it could potentially cause a carry-over effect, provoking the user to have difficulty recalling details from previous prototypes (Charness et al., 2011). This was one of the limitations addressed during the analysis.

In addition, both quantitative (Likert-scale) and qualitative (open-ended questions) were used to receive a more in-depth understanding of the users' reaction towards the prototypes. The mixed methods were a way to gather data from different perspectives.

Furthermore, before conducting any study, ethical considerations were carefully taken into account to avoid any conflicts. Participants were informed beforehand of the structure of the tests. During the study, they were allowed to freely express themselves, creating a more comfortable environment. Users were kept anonymous during the interviews. The only information that was an important part of the test was the participants' age and place of origin in which they were kindly asked to share, as it was a key addition to the data analysis.

## 6 Results

In this section, the results of the user tests are presented. First, the summary results are presented in graphs about the primary metrics, such as the frustration levels, engagement levels, perceived waiting times, and user preferences across the three loading types. Lastly, the emotional feedback of the users will be highlighted.

Thirty participants were recruited to the study from Jönköping University. Their age ranged from 19 to 32 ( $M=22.96$ ,  $SD=3.29$ ). Seven of them identified themselves as male and 23 as female. Their country of origin was diverse from all around the world, including European, Asian, American, and African countries (See in Appendix B).

### 6.1. Frustration Levels

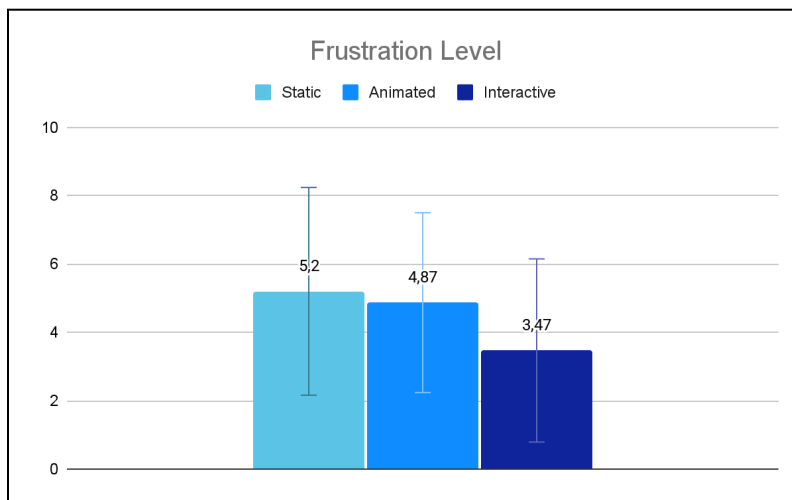


Figure 6.1.1: *Means and Standard Deviations of the Frustration Levels*

As seen in Figure 6.1.1, the mean frustration level for the static screen is the highest, with a mean value of 5.2 ( $SD=3.04$ ). The animated version is slightly less ( $M=4.87$ ,  $SD=2.63$ ), and the participants were least frustrated during the interactive screen ( $M=3.47$ ,  $SD=2.68$ ).



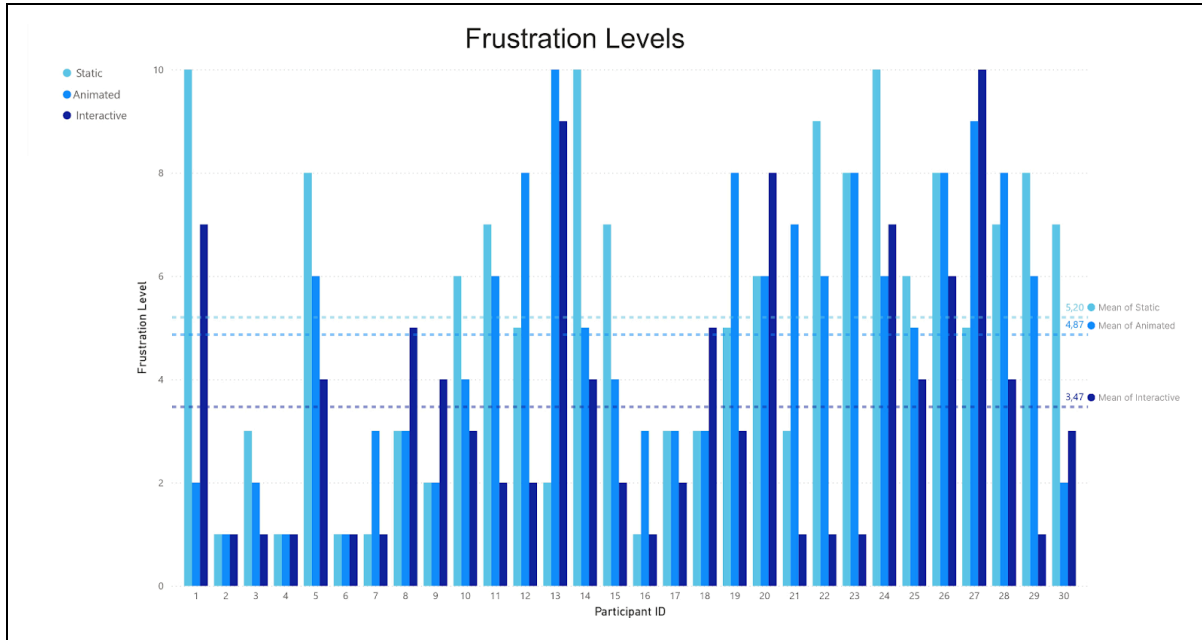


Figure 6.1.2: *Frustration Levels of all IDs*

As seen in Figure 6.1.2, there was a big individual difference in how loading screens affected frustration. This was also reflected in the open-ended questions.

For the **static** screen, the most frequent causes of frustrations mentioned by users were the lack of feedback, movement, which caused boredom, and in some cases, participants associated it with a system failure(frozen screen). One participant said that, “Static made me feel worried, because there was no indication that something was loading”(ID22). Another said, “I was frustrated at a level of 8, because it was not even moving”(ID6).

The **animated** loading screen performed better than the static one in reducing the frustration levels, with a mean of 4.87. A user claimed, “The hypnotising animation calmed me down” (ID17). Furthermore, the movement indicated that something was happening, which decreased the fear of technical issues. Despite the enjoyment of the animation, one of the main sources of frustration was not having enough information about the error.

The most effective loading screen for reducing the frustration levels was the **interactive** one ( $M=3.47$ ). A user stated that, “the more engaging the loading screen is, the less frustrated I get”(ID6). After being engaged with the game, a strong reduction in frustrations appeared because of the distraction from the waiting and emotional uplift. On the other hand, very few participants claimed the interactive game to be rather a cause of frustration. Some of them (ID10) felt the mini-game was "too simple" and wanted more depth, which slightly frustrated them. A user (ID11) also said that the fact that they have to stop playing when the internet comes back

annoys them. Some participants felt a slight frustration because they were unsure how to play in the beginning, or did not even start to play because they did not realise it was a game(ID4).

In **general**, most of the users claimed that the design itself had a positive effect. A user said, “When the loading screen appeared, I felt a frustration level of 11 because of the no connection, but it went down to 3 because the fish was cute”(ID1). On the other hand, few users claimed that the cute design rather made them more frustrated since they could not associate their current annoyed mood caused by the internet connection loss with the happy face of the fish (ID1, 15). Another issue that occurred in all versions was the lack of information. A participant stated, “There are no further options to do, for example, an interactive button for trying to reconnect or information giving more details about the error would be nice”. There was also a user (ID16) who claimed that his frustration level was not primarily influenced by the loading screen but by the moment the loading screen appeared. He would feel that if it appears after buying the shoe, it's just a scam website wanting to steal his money. Hence, his frustration level is more connected to how trusted he feels with the website or the company.

## 6.2. Engagement Levels

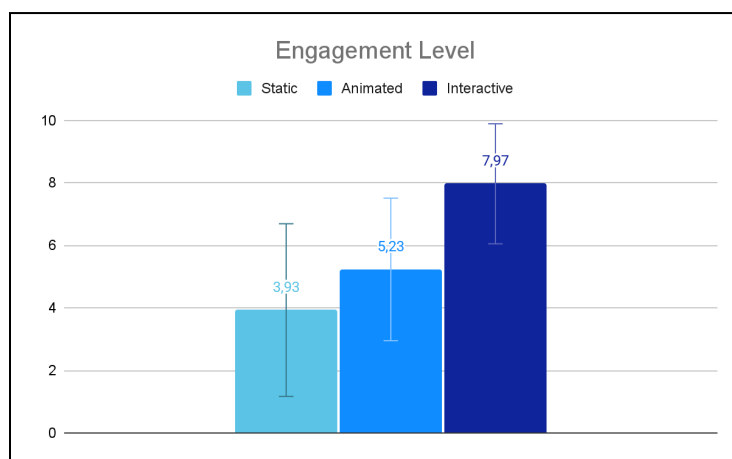


Figure 6.2.1: Means and Standard Deviations of Engagement Levels

The data connected to engagement was measured in the same way as the frustrations. Participants felt least engaged during the static loading screen, on a mean level of 3.93 (SD=2.76)(Figure 6.2.1). The animated screen resulted in a mean of 5.23 (SD=2.28). The highest engagement was present during the interactive screen with the mean value of 7.97 (SD=1.92).

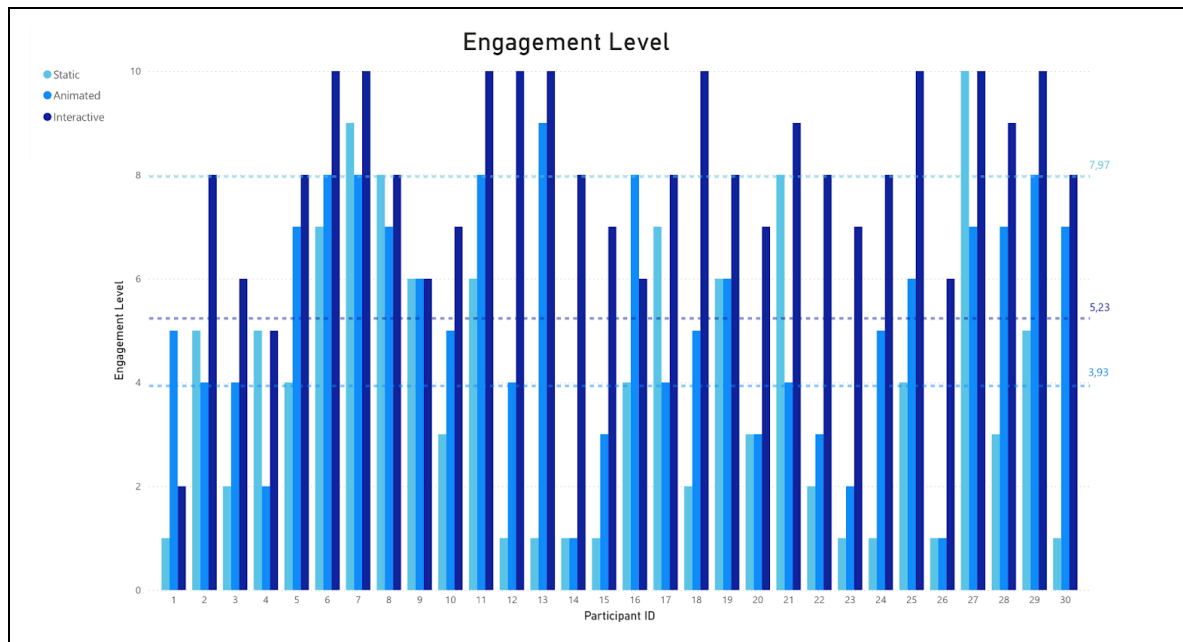


Figure 6.2.2: *Engagement Levels of all IDs*

The engagement level was measured on a 1-10 Likert scale. Here, the difference in mean engagement levels between prototypes was more pronounced than the variation in frustration levels.

The least engagement level was reported for the **static** loading screen ( $M=3.93$ ). The little engagement caused by the static screen was mainly from the appreciation of the visuals. Most of them liked that the design is cute (e.g, ID1, ID2, ID3, ID4, ID5...), and that it is not only a completely blank, white and grey error screen (e.g., ID25). But most of the users claimed this type of screen to be little or not engaging at all, primarily because of the lack of interaction. A user claimed, "I liked the fish, but it was just standing there" (ID6).

Participants found the **animated** screen more engaging than the static one ( $M=5.23$ ). The movement caught their attention, which led to a higher engagement level, saying, "I unintentionally started to count how many times it goes up and down, which drew my attention"(ID17). Many of them mentioned that they were engaged by the animated visuals, like the moving bubbles(e.g, ID6).

The **interactive** loading screen resulted in a far higher engagement level than the rest ( $M=7.97$ ). Almost all users described the interactive screen as "fun", "distracting" (eg, ID8). Active interaction helped users forget about waiting. A user even stated that, "I could at least do something while waiting, gave a new goal"(ID15). Despite creating the highest engagement overall, some negative factors to engagement also occurred. For example, a user who claimed to

play online games frequently did not feel engaged enough because of the simplicity of the game (ID10).

### 6.3. Perceived Time

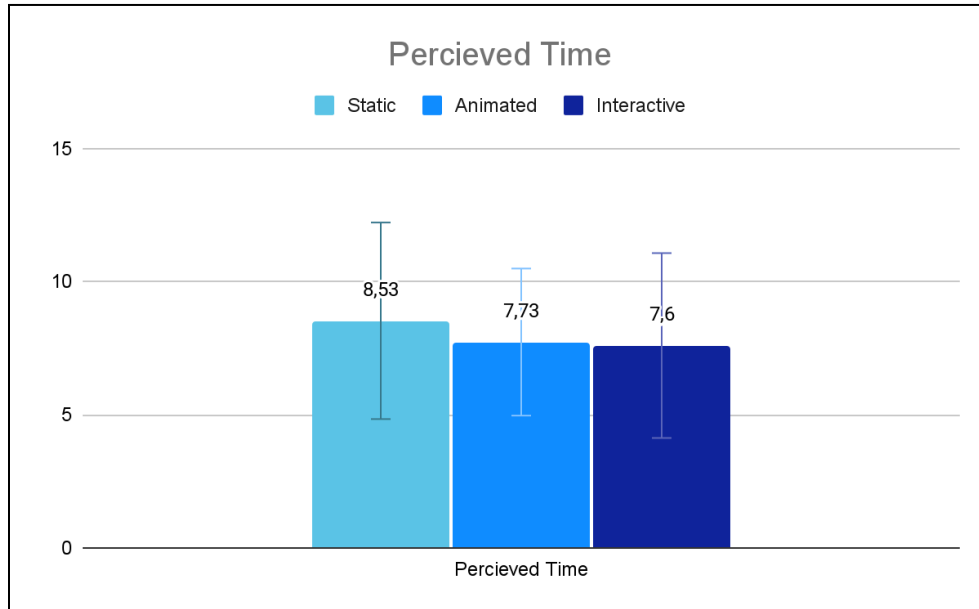


Figure 6.3.1: *Means and Standard Deviations of Perceived Times*

Despite every loading screen lasting 10 seconds, participants perceived the static loading screen to have the longest waiting time ( $M=8.53s$ ,  $SD=3.69$ ). The animated ( $M=7.73s$ ,  $SD=2.76$ ) and the interactive were similarly perceived ( $M=7.60s$ ,  $SD=3.47$ ).

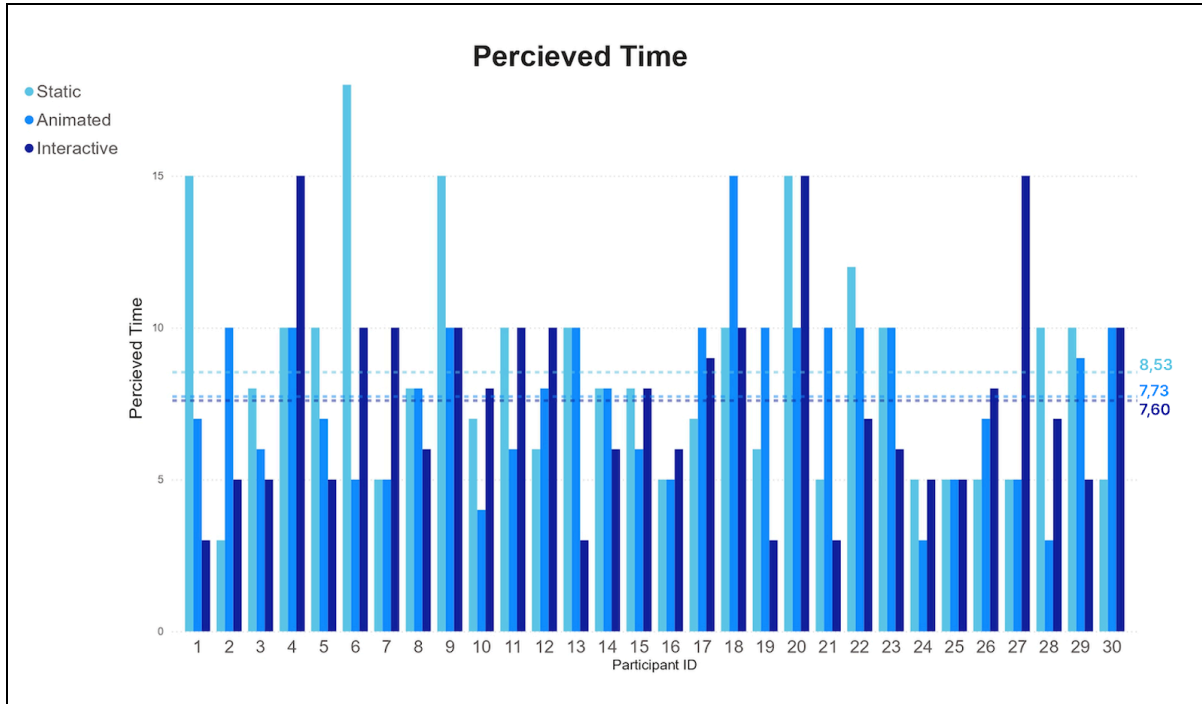


Figure 6.3.2: *Perceived Times of all IDs*

As seen above, in Figure 6.3.2, the perceived waiting times varied widely across users and prototypes, despite all of them lasting ten seconds. Many of the participants claimed that the interface influenced their time perception.

Most of the positive comments were regarding the **Interactive** interface. Many claimed that since they were doing something during the interactive loading screen, not just staring at the screen, the time did not feel long (ID2). A user said, “I could engage with it, which made the waiting time shorter” (ID10). Another claimed, “There were many impulses, I was doing something, so I felt the time was passing (ID1)”. One stated that, “It killed time” (ID14).

For the **static** loading screen, negative comments were more typical: “Nothing was happening, it made the time feel longer (ID22)”. As seen in the graphs, the static prototype resulted in being the longest in terms of the perceived time.

However, one participant uniquely claimed the static loading screen felt the shortest in the general questions stage. He said, “Since it did not capture my attention, I started to do other things, so time passed quicker (ID13)”

The **animated** loading screen mostly resulted in positive comments. At the same time, a user stated that “when it was moving, I felt the time was passing slower (ID13).

Contrary to assuming that the interactive screen will make the time passing feel much faster than the animated or the static, no difference (less than one second, so it is not significant) was found between conditions regarding perceived waiting times. Only eight out of 30 participants stated clearly that the interactive screen lasted longer than the rest of the screen types.

## 6.4. User Preferences

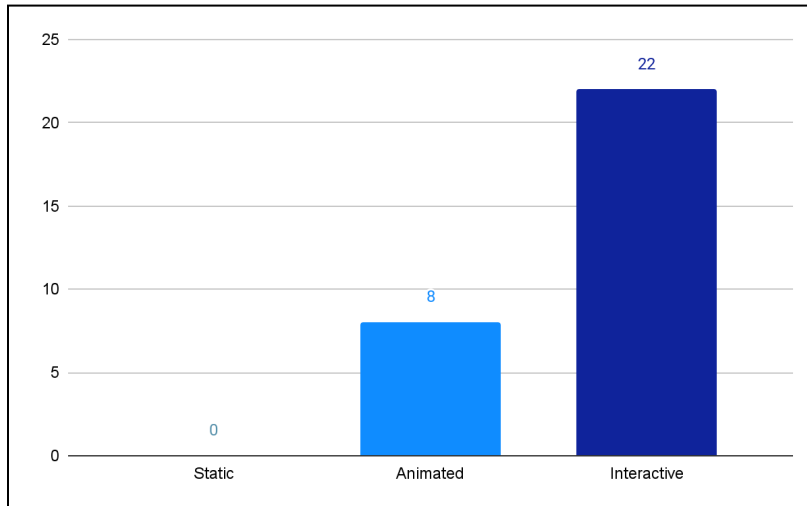


Figure 6.4: *Answers for the overall preference*

Participants were asked which prototype gave them the best experience. Figure 6.4 shows that 22 preferred the interactive loading screens, while the remaining eight chose the animated prototype. Notably, none of the users preferred the static version. The results of the interview questions also indicate a clear preference for the interactive prototype over the others discussed later in more detail.

### 6.4.1. The relationship between the perceived time and frustrations?

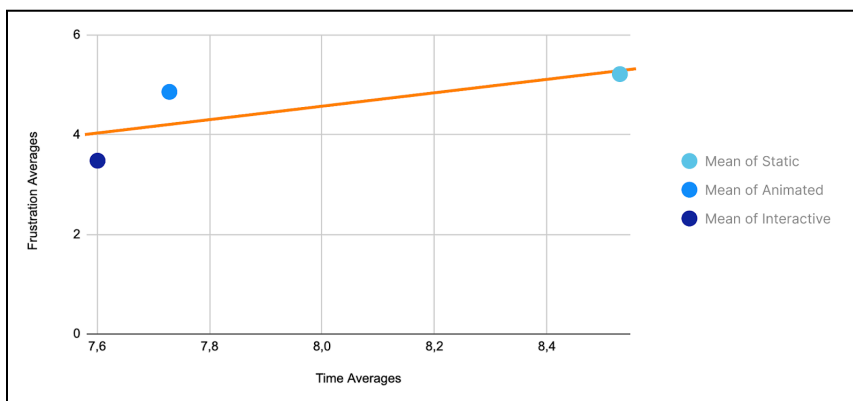


Figure 6.4.2: *Looking for the relation between perceived time and frustrations*

Looking at the data collected, it is visible that when participants perceived time as short, they reported a lower frustration level. This was calculated by using the Pearson correlation to identify the relations between the perceived time and frustration. For the static prototype, it showed a Pearson correlation of  $r=0.06$ , showing no linear relation between perceived time and frustration. Meaning the participants' frustration did not relate to how long the loading took. The results for the animated prototype appeared to have a Pearson correlation of  $r=-0.07$ , indicating a weak negative correlation between the two, explaining that there was a decrease in frustration as the perceived time escalated. As for the interactive prototype, the Pearson correlation of  $r=0.12$  results in a positive yet weak relationship; both variables slightly increased, implying that factors such as the design may have boosted the participants' experience and emotions rather than the perceived time. Stated examples from participants are further proof of frustration, not necessarily having any correlation with the perceived time:

Many participants described their experience and how the game was engaging, which also helped reduce time and frustration. One user preferred *"the GAME, because there was something to do, I did not just stare at the screen"*(ID3), highlighting that engagement lowered perceived time and frustration. Another pointed out *"The GAME, because I got lost in it, I was not bored"*(ID5), which showed a short perceived time linked to minimal frustration. Furthermore, a participant shared that *"My mind was engaged, so I did not focus on how long it was"*(ID8), indicating that time was not the focus, and so frustration was low. There are also cases where long perceived time led to noticeably higher frustration. A user noted, *"It felt very long, it did not blow bubbles"*(ID6), indicating that when there is no distraction, the perceived time feels longer, causing higher frustration. However, another participant said *"STATIC - I could not play, so it felt long"*(ID4), stating that when there is no interaction, the wait feels endless, hence the frustration increases.

This pattern lines up perfectly with cognitive psychology principles. Research shows that this aligns with Yin et al. (2021), who demonstrated that mobile interactivity features such as active control and responsiveness promote cognitive absorption, thereby decreasing users' perception of waiting time and improving their overall satisfaction.

To sum up, the differences in perceived waiting times between conditions were not statistically significant, and no strong correlation was found between perceived waiting time and frustration. Therefore, perceived waiting time does not appear to predict frustration in this study. However, qualitative data (interview responses) shows that when users felt more engaged, they often said the time felt shorter, even if it wasn't shorter. So, while perceived waiting times did not appear to predict frustration, they may be influenced by users' engagement levels.

## **6.5 Emotional Feedback of Prototypes**

During the interviews, participants' responses revealed that each prototype delivered a different kind of feedback and reaction. The static prototype received relatively negative feedback, with users expressing frustration and confusion when nothing was happening during an internet malfunction. This lack of activity left them annoyed and increased their perception of waiting time.

The animation prototype was generally more appealing, as the movement of the design suggested that the internet issue was being addressed. This gave users hope, increasing their attention and engagement with the screen. However, several participants still expressed concern and anxiety about the ongoing situation.

In contrast, the interactive mini-game prototype received mostly positive reactions. Participants expressed how the game was fun and a great way to keep them focused on an enjoyable activity rather than the internet issue. Many found it to be an unexpected yet highly effective way to manage their frustration, making it more engaging than the other prototypes.

Overall, the findings confirmed that the majority of users have a preference for the interactive prototype over the static and animated versions. This suggests that playful, visually engaging designs can have a positive impact on the user, reducing any negative emotion. The interactive prototype delivered both emotional feedback by decreasing frustration and increasing entertainment, and cognitive feedback by making users understand what was happening during the waiting time and the purpose of the prototype during periods of internet malfunctions.



## 7 Discussion

In this study, it was investigated how different types of loading screens—static, animated, and interactive—affect users’ frustration levels, engagement, and perceived waiting times during internet interruptions in an e-commerce context. A within-subjects experimental design was used, involving 30 university participants who interacted with all three prototypes. Data were collected through Likert-scale questions and open-ended interviews immediately after each interaction. The results showed that the interactive loading screen outperformed the static and animated screens in the engagement and frustration levels, but there was no significant difference in perceived time. It reduced frustration, enhanced engagement, and was perceived as having the shortest waiting time. Additionally, 73.3% of participants explicitly stated a preference for the interactive prototype, citing it as fun, distracting, and mood-enhancing. These findings highlight the value of interactivity in user interface design, particularly during periods of forced inactivity such as loading or connectivity issues.

### 7.1. Interpretation of the Results

The consistency between the results of this study and those of Cheng et al. (2024) and Wang et al. (2021) confirms that interactive and visually engaging loading screens notably improve UX by effectively reducing perceived waiting times and enhancing emotional engagement. These collective insights strongly recommend incorporating interactive and thoughtfully designed visual feedback into user interfaces to optimise user satisfaction during unavoidable waiting periods.

In the next sections, the results will be interpreted by the research questions.

#### 7.1.1. RQ1: How does the user interface during long loading times influence the (a) frustration level, (b) engagement level, and (c) perceived time?

The first RQ was split up into three sub-questions: frustration, engagement, and perceived time.

There are several interface factors that influence user **frustration**. After analysing the data, many aspects of the prototypes were found to reduce frustration, while a few elements contributed to it. Participants also provided helpful suggestions for further minimising frustration. Overall, the interactive loading screen proved to be the most effective in reducing frustration during internet reconnection delays.

Regarding **engagement** levels, participants found the static loading screen the least engaging. The animated screen was found more engaging because of the movement. The most engaging was the interactive loading screen because, besides the movements on the screen, users also had the opportunity to interact, which most of them claimed to be “fun”.

The interface had an influence on the **perceived waiting time**, in a different way for each prototype described above. Although the interactive loading screen had the lowest mean perceived waiting time (7.6 seconds), the differences between conditions were not statistically significant.

### **7.1.2. RQ2: Is perceived lower time something that predicts frustration?**

It was hypothesised that lower perceived waiting times would be associated with lower frustration levels, based on prior research indicating that longer perceived delays can intensify negative emotional responses (Heidrich et al., 2020).

Based on the related literature, it was assumed that the interactive loading screen would provide the shortest time perception. However, the quantitative findings did not support this theory. The Pearson correlation coefficients between perceived time and frustration were weak across all three prototypes (static:  $r = 0.06$ ; animated:  $r = -0.07$ ; interactive:  $r = 0.12$ ), indicating no statistically significant relationship. This suggests that perceived wait time, in isolation, did not meaningfully predict the level of user frustration.

In contrast, the qualitative data provided additional insight. Many participants described the interactive screen as more enjoyable and less frustrating, not necessarily because the wait felt shorter, but because they were actively engaged. Statements such as “I got lost in the game” and “I didn’t think about the time” imply that engagement may have influenced their perception of time and emotional response.

These results suggest that while perceived waiting time did not predict frustration in a measurable way, engagement may be a more relevant factor in shaping both time perception and emotional outcomes during waiting experiences.

### **7.1.3. RQ3: Which type of interface (static, animated, interactive) has the most positive effect on user’s experience during long loading times?**

This broader research question serves as a good conclusion and shows the most important goal of the whole study. For answering, the quantitative and qualitative results can be used as well.

First of all, the measurement of the frustration and engagement levels is relevant to look at. The prototype, which has the lowest frustration level and the highest engagement level, provides the most positive experience. As stated earlier, this prototype is the **interactive** loading screen.

Furthermore, as found in the RQ3, shorter time perception means less frustration, hence, a better experience. The mean value for the perceived waiting times for the loading screens was shortest

for the **interactive** loading screen. Besides the collected numbers, asking the opinion of the users was also a relevant factor for this research question. In the post-test interview, a question was asked: “*Which loading screen prototype gave you the best experience and why?*”. 22 participants out of 30, so 73,3% claimed that the **interactive** loading screen gave them the best experience. They cited key reasons such as its effectiveness in distracting them from the wait, engaging nature, entertainment value, mood enhancement, and ability to make time feel as though it passed more quickly..

To sum up, for long waiting times (10s), the interactive loading screen performed best, so among these three prototypes, the interactive one serves the best UX.

## 7.2. Real World Implications

Internet interruptions and connectivity issues are common occurrences in daily online experiences. Situations such as poor connection and long waiting times can enhance the user’s frustration, making loading screens a crucial element of user satisfaction. Therefore, they should be carefully considered during website development.

When participants were asked, “*In a real-life situation, which of the three prototypes would you like to see?*”, the majority preferred having an interactive screen, which allowed them to engage in an activity while waiting. It is also suggested that as the waiting time increases, the interactive loading screen may be preferred even more.

Even though the animated prototype was also favoured by some users, it didn't fully meet their need for engaging. The majority argued that both the static and animated were “*cute but boring*” or noted that the “*simple movement*” still reminded them of the waiting process. In contrast, the interactive prototype redirected the users’ attention to the mini-game, shifting their focus away from the internet problem. Many expressed that “*time passed faster; I forgot about waiting*” and that they “*liked doing something active instead of just waiting*”.

Since user satisfaction can influence brands’ reputation (Walsh et al., 2009), it is important that companies care about all aspects of the UX, including longer waiting times. Our finding suggests that companies and designers should prioritise user satisfaction when designing a website, as this can have an impact on the users’ perception of the brand, especially during internet interruptions. As previously mentioned, Faisal et al. (2017) suggest having a well-designed site in order to make the users trust the company and have a more engaging and satisfying experience. While static and animated screens may be easier and quicker to implement, an interactive screen can have a more positive impact on UX and brand trust.

### 7.3. Limitations & Future Work

This section outlines the key limitations again, but with new added encounters during the study as well as new approaches for future research to address these challenges. Furthermore, inspired by the findings after the study, further research ideas were born, which will be described at the end of this section.

#### 7.3.1. Carryover Effect

As mentioned in the earlier section, the mean of the time perception for the static loading screen was 8.53 seconds when considering all 30 participants. However, knowing the carryover effect described in the “Scope and Limitations” section, it is known that the results can be affected in the within-subjects design. Comparing the users who have seen the static prototype as first, second, and third can also prove this issue to be true.

As seen in the Figure below, the 10 participants, who had seen the static loading screen as the first prototype, perceived the loading screen to be a mean of 6.7 seconds long. The mean time is 9 seconds for the participants seeing it as the second prototype, and it is 10 seconds for the ones seeing it as the last prototype.



Figure 7.3.1: Time Perceptions for Static Loading Screen

Even the comments of the participants prove this issue. For example, for the question “In your opinion, which loading screen made the wait feel the shortest? Why?” participant-7 answered: “I felt the STATIC was the shortest, maybe because it was the first prototype and I was not bored of it yet”(ID7).

To reduce this risk, future research could implement a between-subjects design, assigning participants to interact with only one loading screen. This approach would help isolate the impact of each prototype by removing inter-prototype comparisons.

### 7.3.2. Recall Accuracy

Recall accuracy is also a factor that can influence the results. In Figure 7.3.2, the perceived waiting times of the users are presented, which were measured after each prototype. After seeing all three prototypes, the question was asked of the participants, “Which loading screen made the wait feel the shortest?” 5 participants (16,67%) answered this question not with the loading screen they actually perceived as the shortest when asked right after each prototype. For example, Participant-26 answered at the end that she felt the interactive loading screen was the shortest; however, earlier she claimed it to be the longest, 8 seconds, besides claiming the animation to be 7 seconds and the static to be 5 seconds.

| Participant ID | Perceived Time<br>(asked after each prototype) |          |             | Which loading screen<br>made the wait feel the<br>shortest?<br>(Asked after all the prototypes) |
|----------------|------------------------------------------------|----------|-------------|-------------------------------------------------------------------------------------------------|
|                | STATIC                                         | ANIMATED | INTERACTIVE |                                                                                                 |
| 30             | 5                                              | 10       | 10          | Interactive                                                                                     |
| 26             | 5                                              | 7        | 8           | Interactive                                                                                     |
| 15             | 8                                              | 6        | 8           | Interactive                                                                                     |
| 28             | 10                                             | 3        | 7           | Interactive                                                                                     |
| 13             | 10                                             | 10       | 3           | Static                                                                                          |

Figure 7.3.2: Comparing the perceived time asked right after each prototype and at the end of the whole interview

This undermines the reliability of self-reported perceived time, as participants' responses did not always align between the immediate and final evaluations, making interpretation more challenging.

To address this in future research, capturing even more immediate reactions could be implemented. For example, asking, “How long do you think that took?” right after each loading screen appears, not after the prototype.

### **7.3.3. Wording Confusion in the Interview Questions**

Some parts of the wording in the post-test interview were sometimes unclear for the users. Terms such as ‘loading screen’ and ‘engagement’ required clarification during interviews. After researchers paraphrased “loading-screen” into “no-internet connection screen” or “error-message screen,” users seemed to understand them better.

Ambiguity in question wording can affect the validity of qualitative feedback, especially if participants interpret key terms differently. To avoid this issue, future studies should consider pilot-testing interview questions with a small sample before the full experiment, ensuring that terminology is intuitive and clearly understood by participants.

### **7.3.4. Limited Participant demographics**

The participant sample consisted entirely of university students aged 19–32. While this group offered useful feedback, the results cannot be generalised to broader populations with varying ages, cultures, technological familiarity, or preferences. This limits the external validity of the findings, particularly for older or less digitally experienced user groups who may experience loading screens differently.

Future research should aim to broaden the demographic scope, including participants from different age groups, professions, and digital skill levels, to gain more comprehensive insights into user experience across diverse populations.

### **7.3.5. Suggestions for Further Research**

The findings of this study provide valuable insights into the impact of static, animated, and interactive loading screens on user frustration, engagement, and perceived waiting time. However, they also open several interesting areas for further research.

Future studies could also explore **a broader range of interactive designs**. The mini-game used in this research was intentionally simple to ensure ease of use, but more complex or varied forms of interaction could be tested to examine whether different types of engagement (e.g., quizzes, puzzles, storytelling) have an even greater impact on user satisfaction and perceived wait times. It would be particularly interesting to investigate the level of complexity to optimise user experiences even further.

Finally, designing the **optimal loading** screen would be an interesting further study. Participants during the user tests shared valuable suggestions and thought about how they think the screens could be even better. These included opinions about colours, movements, characters, backgrounds, vibes, different types of games and many more creative ideas. To mention some examples, one user (ID6) suggested having “more stimulations” on the screen, for example, more objects like corrals in the background”, another (ID7) mentioned a dolphin as a character. Regarding the text, a user (ID9) suggested a more serious font and black colour. A user (ID15) suggested having a game in the same theme as the website, so, for example, broken, old shoes falling from the sky, which the user needs to avoid, and discount icons to catch for raising the score. Some users (ID1, 15) suggested having a character whom they can emotionally associate with, and being angry because of the internet loss. Some users (ID1) even suggested a whole animated “story” to appear on the screen, a character who is having a BBQ, and instead of marshmallows, burning their nerve cells because of the frustration caused by the lack of internet. A user (ID15) said that involving the text “no internet” somehow in the movement would make him notice it more, because during the interactive loading screen, he was focusing on the game and did not pay attention to the error message. It could be solved, for example, if the letters of the “No Internet Connection” were to fall down, and those would be the objects to catch for the scores.

## 8 Conclusion

The purpose of this study was to explore the effect of three loading screen types - static, animated, and interactive - on user satisfaction, frustration, and engagement during long waiting times in an e-commerce website. There is limited research directly comparing these specific types of loading screens, and about the importance of UX during internet interruptions appearing on websites. Thus, this research set out to test those issues and gaps by evaluating users' responses to the prototypes in a controlled setting.

The main objective of this research was to identify how the user interface can influence the three main findings of UX: levels of frustration, engagement, and the perception of time. More specifically, it demands whether perceived time can predict frustration, and which prototype was most effective during long waiting times. In response to that, a controlled study was conducted with repeated measures, with a total of 30 participants aged 19-32. They were assigned to interact with three loading screens in a randomised order and share their experience and preferences. Likert scales and open-ended questions were used as quantitative methods.

The results revealed a strong user preference (73.3%) for the interactive screen as opposed to the static and animated. It states that there was a difference between frustration and engagement level, yet the perceived time remained relatively consistent. A qualitative method was also used for this research. Participants were asked general questions, and went more in-depth about their experience. Moreover, users' feedback allowed a clearer understanding of why the interactive screen was most favoured. It made the experience "more enjoyable" while reducing frustration, as well as making the time perception feel shorter. This supports previous research done by Cheng et al. (2024) and Wang et al. (2021) that confirm interactive screens enhance UX by improving user engagement and lower the perceived waiting time.

With these findings, this study strongly recommends implementing loading screens on websites, as it has a positive effect on UX. In particular, interactive screens with well-designed features that can grab users' attention, and colours that are carefully chosen to evoke emotional feedback. It is important to note that the loading screen can provide user satisfaction as well as build trust with the customer, benefiting the company and its image.

In conclusion, the research highlights the importance of UX and how loading screens can enhance user satisfaction, especially during waiting times. Loading screens can increasingly offer a positive outcome to the user, as they play a vital role in developing emotional feedback and user perception.



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## 10 Appendixes

Appendix A Post-Test Interview Answers

Appendix B Table of Participant Information

## **Appendix A**

Link to the Post-Test Interview Answers of the participants:

<https://docs.google.com/document/d/1GXnObktRZAqyk1ISx5UAI7dTFkdr2N5fmFecVQnjvkA/edit?tab=t.0#heading=h.h54qn6ujm4n1>

## Appendix B

### Participant Information

| Participant ID | Origin         | Gender | Age |
|----------------|----------------|--------|-----|
| 1              | Hungary        | female | 22  |
| 2              | Hungary        | female | 23  |
| 3              | Hungary        | female | 21  |
| 4              | Hungary        | female | 30  |
| 5              | Hungary        | female | 19  |
| 6              | Hungary        | female | 19  |
| 7              | Hungary        | female | 24  |
| 8              | Ghana          | female | 22  |
| 9              | Turkey/Germany | female | 19  |
| 10             | Croatia        | male   | 22  |
| 11             | Sweden         | female | 24  |
| 12             | Czech Republic | female | 21  |
| 13             | Czech Republic | male   | 23  |
| 14             | Czech Republic | male   | 24  |
| 15             | Hungary        | male   | 26  |
| 16             | Poland         | male   | 19  |
| 17             | Hungary        | female | 21  |
| 18             | Cyprus         | male   | 24  |
| 19             | Italy          | female | 25  |
| 20             | Honduras       | female | 21  |
| 21             | Japan          | female | 31  |
| 22             | Brazil         | female | 32  |
| 23             | Italy          | female | 22  |
| 24             | USA            | female | 24  |
| 25             | Ukraine        | female | 20  |

|    |          |        |    |
|----|----------|--------|----|
| 26 | Slovakia | female | 22 |
| 27 | Iran     | male   | 24 |
| 28 | Germany  | female | 21 |
| 29 | Romania  | female | 22 |
| 30 | Estonia  | female | 22 |